

# Mathera

## Complete Curriculum Map

1132 skills

7 eras · 87 domains / branches

| Era   | Name    | World        | Domains | Skills |
|-------|---------|--------------|---------|--------|
| I     | Count   | First Grove  | 7       | 108    |
| II    | Operate | Forge City   | 10      | 135    |
| III   | Relate  | Bridge Isles | 9       | 146    |
| IV    | Solve   | Peak Range   | 15      | 199    |
| V     | Prove   | Proof Palace | 15      | 173    |
| VI    | Motion  | Tide Sea     | 12      | 174    |
| VII   | Space   | Star Vault   | 19      | 197    |
| TOTAL |         |              | 87      | 1132   |

Every skill below is listed Era -> Domain -> Skill. Eras I-III from your design lists (Era II now complete through II.10). Eras IV-VII follow the v2 outline.

## Contents by domain

### Era I Count (108 skills)

|     |                        |    |
|-----|------------------------|----|
| I.1 | Counting               | 16 |
| I.2 | Comparing & Ordering   | 12 |
| I.3 | Place Value            | 12 |
| I.4 | Addition & Subtraction | 23 |
| I.5 | Patterns & Logic       | 15 |
| I.6 | Shape & Space          | 14 |
| I.7 | Measure & Data         | 16 |

### Era II Operate (135 skills)

|       |                         |    |
|-------|-------------------------|----|
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| II.2  | Multiplication          | 15 |
| II.3  | Division                | 15 |
| II.4  | Factors & Multiples     | 12 |
| II.5  | Fractions I             | 15 |
| II.6  | Fractions II            | 14 |
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| II.8  | Expressions             | 9  |
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### Era III Relate (146 skills)

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| III.4 | Exponents & Roots        | 17 |
| III.5 | Expressions              | 15 |
| III.6 | Equations & Inequalities | 12 |
| III.7 | Lines & First Functions  | 12 |
| III.8 | Geometry                 | 23 |
| III.9 | Statistics & Probability | 20 |

### Era IV Solve (199 skills)

|       |                              |    |
|-------|------------------------------|----|
| IV.1  | Equations & inequalities     | 5  |
| IV.2  | Functions                    | 15 |
| IV.3  | Linear functions             | 9  |
| IV.4  | Systems                      | 15 |
| IV.5  | Exponents & radicals         | 7  |
| IV.6  | Polynomials                  | 16 |
| IV.7  | Quadratics                   | 17 |
| IV.8  | Complex numbers              | 16 |
| IV.9  | Function toolkit             | 18 |
| IV.10 | Polynomial functions         | 14 |
| IV.11 | Rational & radical functions | 14 |
| IV.12 | Exponential & logarithmic    | 17 |
| IV.13 | Sequences & series           | 14 |
| IV.14 | Counting & probability       | 12 |
| IV.15 | Data & distributions         | 10 |

### Era V Prove (173 skills)

|     |                |    |
|-----|----------------|----|
| V.1 | Logic & proof  | 16 |
| V.2 | Lines & angles | 6  |

|      |                        |    |
|------|------------------------|----|
| V.3  | Triangles              | 14 |
| V.4  | Polygons               | 9  |
| V.5  | Similarity             | 12 |
| V.6  | Right-triangle trig    | 14 |
| V.7  | General triangles      | 9  |
| V.8  | Circles                | 17 |
| V.9  | Trig functions         | 15 |
| V.10 | Identities & equations | 12 |
| V.11 | Coordinate geometry    | 9  |
| V.12 | Conic sections         | 11 |
| V.13 | Transformations        | 12 |
| V.14 | Solids                 | 8  |
| V.15 | Constructions          | 9  |

## Era VI Motion (174 skills)

|       |                             |    |
|-------|-----------------------------|----|
| VI.1  | Limits & continuity         | 20 |
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| VI.5  | Integration techniques      | 16 |
| VI.6  | Applications of integrals   | 15 |
| VI.7  | Differential equations I    | 12 |
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| VI.9  | Sequences & series          | 11 |
| VI.10 | Power & Taylor series       | 16 |
| VI.11 | Statistical inference       | 9  |
| VI.12 | Mechanics                   | 9  |

## Era VII Space (197 skills)

|        |                                |    |
|--------|--------------------------------|----|
| VII.1  | Vectors in $\mathbb{R}^n$      | 12 |
| VII.2  | Linear systems                 | 9  |
| VII.3  | Matrix algebra                 | 12 |
| VII.4  | Vector spaces                  | 16 |
| VII.5  | Linear transformations         | 9  |
| VII.6  | Determinants                   | 9  |
| VII.7  | Eigenvalues & eigenvectors     | 9  |
| VII.8  | Orthogonality                  | 12 |
| VII.9  | Symmetric matrices & SVD       | 8  |
| VII.10 | Functions of several variables | 6  |
| VII.11 | Partial derivatives            | 12 |
| VII.12 | Optimization                   | 8  |
| VII.13 | Multiple integrals             | 9  |
| VII.14 | Vector calculus                | 18 |
| VII.15 | First-order equations          | 7  |
| VII.16 | Second-order linear            | 13 |
| VII.17 | Systems of ODEs                | 11 |
| VII.18 | Laplace transform              | 9  |
| VII.19 | Fourier series                 | 8  |

# Era I . Count - First Grove

108 skills . 7 domains . Things begin here.

## I.1 . COUNTING (16)

---

- 1.01 Count objects one by one
- 1.02 Number words in order
- 1.03 Count to 20
- 1.04 Count to 50
- 1.05 Count to 120
- 1.06 Read and write numerals to 20
- 1.07 How many in all
- 1.08 Zero
- 1.09 See amounts to 5 at a glance
- 1.10 See amounts to 10 at a glance
- 1.11 Count on
- 1.12 Count back
- 1.13 One more, one less
- 1.14 Skip count by 10s
- 1.15 Skip count by 5s
- 1.16 Skip count by 2s

## I.2 . COMPARING & ORDERING (12)

---

- 2.01 More or fewer by matching
- 2.02 Equal groups
- 2.03 Number line 0-10
- 2.04 Compare numbers to 10
- 2.05 Number line 0-20
- 2.06 Order numbers
- 2.07 Ordinal numbers
- 2.08 Compare to 20
- 2.09 Compare to 100
- 2.10 Symbols =, > and <
- 2.11 Estimate a quantity
- 2.12 How many more, how many fewer

## I.3 . PLACE VALUE (12)

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- 3.01 Ten ones make a ten
- 3.02 Teen numbers are a ten and some ones
- 3.03 Tens and ones to 50
- 3.04 Tens and ones to 99
- 3.05 Bundle and unbundle
- 3.06 Hundreds
- 3.07 Three-digit place value
- 3.08 Expanded form
- 3.09 Number names
- 3.10 10 more, 10 less
- 3.11 The hundred chart
- 3.12 Compare using place value

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**I.4 . ADDITION & SUBTRACTION (23)**

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|      |                                   |
|------|-----------------------------------|
| 4.01 | Put together                      |
| 4.02 | Take away                         |
| 4.03 | Addition stories                  |
| 4.04 | Subtraction stories               |
| 4.05 | Part-part-whole                   |
| 4.06 | Number bonds to 5                 |
| 4.07 | Number bonds to 10                |
| 4.08 | Doubles                           |
| 4.09 | Near doubles                      |
| 4.10 | Make ten                          |
| 4.11 | Count on to add                   |
| 4.12 | Add in any order                  |
| 4.13 | Subtract by counting back         |
| 4.14 | Subtract by counting up           |
| 4.15 | Subtract through ten              |
| 4.16 | Fact families                     |
| 4.17 | Missing addends                   |
| 4.18 | The equal sign as a balance       |
| 4.19 | Addition facts within 20          |
| 4.20 | Subtraction facts within 20       |
| 4.21 | Add and subtract tens             |
| 4.22 | Two-digit addition with models    |
| 4.23 | Two-digit subtraction with models |

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**I.5 . PATTERNS & LOGIC (15)**

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|      |                            |
|------|----------------------------|
| 5.01 | Copy a pattern             |
| 5.02 | Extend a repeating pattern |
| 5.03 | Patterns and rules         |
| 5.04 | Growing patterns           |
| 5.05 | Number patterns            |
| 5.06 | Sort by one attribute      |
| 5.07 | Sort by two attributes     |
| 5.08 | Odd and even               |
| 5.09 | Same and different         |
| 5.10 | True or false              |
| 5.11 | Always, sometimes, never   |
| 5.12 | If-then                    |
| 5.13 | Logic grid puzzles         |
| 5.14 | Order by clues             |
| 5.15 | Guess my number            |

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**I.6 . SHAPE & SPACE (14)**

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|      |                        |
|------|------------------------|
| 6.01 | Name flat shapes       |
| 6.02 | Shape features         |
| 6.03 | Shapes in any position |
| 6.04 | Name solid shapes      |
| 6.05 | Solid shape features   |
| 6.06 | Put shapes together    |
| 6.07 | Take shapes apart      |
| 6.08 | Solids from blocks     |

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- |      |                      |
|------|----------------------|
| 6.09 | Position words       |
| 6.10 | Directions on a grid |
| 6.11 | Halves               |
| 6.12 | Quarters             |
| 6.13 | Symmetry             |
| 6.14 | Shape riddles        |

## **I.7 . MEASURE & DATA (16)**

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- |      |                       |
|------|-----------------------|
| 7.01 | Compare lengths       |
| 7.02 | Order by length       |
| 7.03 | Nonstandard units     |
| 7.04 | Measure with a ruler  |
| 7.05 | Compare weights       |
| 7.06 | Compare capacity      |
| 7.07 | Time to the hour      |
| 7.08 | Time to the half hour |
| 7.09 | Days and months       |
| 7.10 | Read a calendar       |
| 7.11 | Coins                 |
| 7.12 | Make amounts          |
| 7.13 | Tally marks           |
| 7.14 | Picture graphs        |
| 7.15 | Bar graphs            |
| 7.16 | Ask with data         |

## Era II . Operate - Forge City

135 skills . 10 domains . Do things to numbers.

### II.1 . PLACE VALUE TO MILLIONS (12)

---

- 1.01 Thousands
- 1.02 Numbers to a million
- 1.03 Place value patterns
- 1.04 Compare and order large numbers
- 1.05 Round to tens and hundreds
- 1.06 Round large numbers
- 1.07 Estimate sums and differences
- 1.08 Column addition, no regrouping
- 1.09 Column addition with regrouping
- 1.10 Column subtraction, no regrouping
- 1.11 Column subtraction with regrouping
- 1.12 Addition and subtraction problems

### II.2 . MULTIPLICATION (15)

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- 2.01 Equal groups
- 2.02 Arrays
- 2.03 Multiply by 0, 1, 2, 5 and 10
- 2.04 Facts for 3, 4 and 6
- 2.05 Facts for 7, 8 and 9
- 2.06 All facts to 10 x 10
- 2.07 The area model
- 2.08 Properties of multiplication
- 2.09 Multiply by multiples of 10
- 2.10 Two-digit by one-digit
- 2.11 Three- and four-digit by one-digit
- 2.12 Two-digit by two-digit
- 2.13 Multi-digit multiplication
- 2.14 Multiplication problems
- 2.15 Multiplication patterns

### II.3 . DIVISION (15)

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- 3.01 Sharing equally
- 3.02 Making groups
- 3.03 Division as a missing factor
- 3.04 Division facts
- 3.05 Dividing with 0 and 1
- 3.06 Remainders
- 3.07 Interpreting remainders
- 3.08 Divide multiples of 10
- 3.09 Two-digit by one-digit division
- 3.10 Long division by one digit
- 3.11 Division by two digits
- 3.12 Divisibility rules
- 3.13 Division problems
- 3.14 Mixed operations
- 3.15 Estimating quotients

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**II.4 . FACTORS & MULTIPLES (12)**

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|      |                             |
|------|-----------------------------|
| 4.01 | Factor pairs                |
| 4.02 | Multiples                   |
| 4.03 | Primes and composites       |
| 4.04 | The sieve of Eratosthenes   |
| 4.05 | Prime factorization         |
| 4.06 | Common factors              |
| 4.07 | Using the GCF               |
| 4.08 | Common multiples            |
| 4.09 | Using the LCM               |
| 4.10 | Square numbers              |
| 4.11 | Even and odd rules          |
| 4.12 | Number puzzles with factors |

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**II.5 . FRACTIONS I (15)**

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|      |                                  |
|------|----------------------------------|
| 5.01 | Equal parts                      |
| 5.02 | Unit fractions                   |
| 5.03 | Fractions of a whole             |
| 5.04 | Fractions of a set               |
| 5.05 | Fractions on a number line       |
| 5.06 | Whole numbers as fractions       |
| 5.07 | Equivalent fractions with models |
| 5.08 | Equivalent fractions by rule     |
| 5.09 | Compare, same denominator        |
| 5.10 | Compare, same numerator          |
| 5.11 | Compare any fractions            |
| 5.12 | Mixed numbers                    |
| 5.13 | Improper fractions               |
| 5.14 | Fractions as division            |
| 5.15 | Fraction sense                   |

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**II.6 . FRACTIONS II (14)**

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|      |                                    |
|------|------------------------------------|
| 6.01 | Add, like denominators             |
| 6.02 | Subtract, like denominators        |
| 6.03 | Break fractions apart              |
| 6.04 | Mixed numbers, like denominators   |
| 6.05 | Common denominators                |
| 6.06 | Add, unlike denominators           |
| 6.07 | Subtract, unlike denominators      |
| 6.08 | Mixed numbers, unlike denominators |
| 6.09 | Fraction x whole number            |
| 6.10 | Multiply fractions                 |
| 6.11 | Multiplying as scaling             |
| 6.12 | Unit fraction / whole number       |
| 6.13 | Whole number / unit fraction       |
| 6.14 | Fraction problems                  |

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**II.7 . DECIMALS (14)**

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|      |                             |
|------|-----------------------------|
| 7.01 | Tenths                      |
| 7.02 | Hundredths                  |
| 7.03 | Decimal place value         |
| 7.04 | Fractions and decimals      |
| 7.05 | Compare decimals            |
| 7.06 | Round decimals              |
| 7.07 | Add decimals                |
| 7.08 | Subtract decimals           |
| 7.09 | Powers of ten and the point |
| 7.10 | Decimal x whole number      |
| 7.11 | Multiply decimals           |
| 7.12 | Decimal / whole number      |
| 7.13 | Divide by a decimal         |
| 7.14 | Money                       |

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**II.8 . EXPRESSIONS (9)**

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|      |                             |
|------|-----------------------------|
| 8.01 | Order of operations         |
| 8.02 | Write numerical expressions |
| 8.03 | Evaluate expressions        |
| 8.04 | Number patterns and rules   |
| 8.05 | Input-output tables         |
| 8.06 | Missing numbers             |
| 8.07 | Equivalent expressions      |
| 8.08 | Problems to expressions     |
| 8.09 | Balance puzzles             |

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**II.9 . MEASUREMENT & GEOMETRY (20)**

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|      |                          |
|------|--------------------------|
| 9.01 | Metric length            |
| 9.02 | Imperial length          |
| 9.03 | Mass                     |
| 9.04 | Capacity                 |
| 9.05 | Time                     |
| 9.06 | Perimeter                |
| 9.07 | Area by counting         |
| 9.08 | Area of rectangles       |
| 9.09 | Perimeter vs area        |
| 9.10 | Volume by counting cubes |
| 9.11 | Volume of boxes          |
| 9.12 | Points, lines and rays   |
| 9.13 | Angles                   |
| 9.14 | Measure angles           |
| 9.15 | Add angles               |
| 9.16 | Classify triangles       |
| 9.17 | Classify quadrilaterals  |
| 9.18 | Lines of symmetry        |
| 9.19 | The first quadrant       |
| 9.20 | Measurement problems     |

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**II.10 . DATA (9)**

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- 10.01 Line plots
- 10.02 Bar graphs with scales
- 10.03 Pictographs with keys
- 10.04 Tables and frequency
- 10.05 The mean as a fair share
- 10.06 A typical value
- 10.07 Line graphs
- 10.08 Collect and organize data
- 10.09 Read graphs critically

# Era III . Relate - Bridge Isles

146 skills . 9 domains . How quantities connect.

## III.1 . INTEGERS & RATIONALS (15)

---

- 1.01 Negative numbers in context
- 1.02 Integers on a number line
- 1.03 Absolute value
- 1.04 Add integers
- 1.05 Subtract integers
- 1.06 Multiply integers
- 1.07 Divide integers
- 1.08 Order of operations with integers
- 1.09 Rational numbers
- 1.10 Add and subtract rationals
- 1.11 Multiply and divide rationals
- 1.12 Terminating and repeating decimals
- 1.13 Repeating decimals to fractions
- 1.14 Compare and order rationals
- 1.15 All four quadrants

## III.2 . RATIOS & RATES (15)

---

- 2.01 Ratio language
- 2.02 Equivalent ratios
- 2.03 Ratio tables
- 2.04 Unit rates
- 2.05 Rates with fractions
- 2.06 Ratios on a graph
- 2.07 Proportions
- 2.08 Proportional relationships
- 2.09 The constant of proportionality
- 2.10 Scale drawings
- 2.11 Unit conversion with ratios
- 2.12 Sharing in a ratio
- 2.13 Speed, distance, time
- 2.14 Comparing ratios
- 2.15 Ratio puzzles

## III.3 . PERCENT (17)

---

- 3.01 Percent means per hundred
- 3.02 Fractions, decimals, percents
- 3.03 Percent of a number
- 3.04 Find the whole
- 3.05 Find the percent
- 3.06 Over 100% and under 1%
- 3.07 Percent increase
- 3.08 Percent decrease
- 3.09 Discounts and sale prices
- 3.10 Tax and tips
- 3.11 Markup and profit
- 3.12 Simple interest

- 3.13 Percent error
- 3.14 Percent change traps
- 3.15 Percent in data
- 3.16 Estimating with percents
- 3.17 Multi-step percent problems

### III.4 . EXPONENTS & ROOTS (17)

---

- 4.01 Whole-number exponents
- 4.02 Powers of ten
- 4.03 Exponents in the order of operations
- 4.04 Multiplying powers
- 4.05 Dividing powers
- 4.06 Power of a power
- 4.07 Negative exponents
- 4.08 Scientific notation
- 4.09 Calculating in scientific notation
- 4.10 Comparing huge and tiny numbers
- 4.11 Square roots
- 4.12 Cube roots
- 4.13 Estimate roots
- 4.14 Irrational numbers
- 4.15 The real number system
- 4.16 Compare irrationals
- 4.17 Growth with exponents

### III.5 . EXPRESSIONS (15)

---

- 5.01 Variables
- 5.02 Expressions from words
- 5.03 Evaluate expressions
- 5.04 Parts of an expression
- 5.05 Like terms
- 5.06 The distributive property
- 5.07 Factor out a GCF
- 5.08 Equivalent expressions
- 5.09 Simplify expressions
- 5.10 Expressions in geometry
- 5.11 Expressions in context
- 5.12 Expressions with powers
- 5.13 Patterns to expressions
- 5.14 Algebra tiles
- 5.15 Why number tricks work

### III.6 . EQUATIONS & INEQUALITIES (12)

---

- 6.01 What an equation is
- 6.02 One-step equations: + and -
- 6.03 One-step equations: x and /
- 6.04 Two-step equations
- 6.05 Multi-step equations
- 6.06 Variables on both sides
- 6.07 Equations from word problems
- 6.08 Inequalities

- 6.09 One-step inequalities
- 6.10 Two-step inequalities
- 6.11 Rearrange formulas
- 6.12 Choosing a strategy

### III.7 . LINES & FIRST FUNCTIONS (12)

---

- 7.01 Input-output rules
- 7.02 Graph from a table
- 7.03 Rate of change
- 7.04 Slope
- 7.05 Slope and similar triangles
- 7.06  $y = mx + b$
- 7.07 Proportional vs non-proportional
- 7.08 Linear vs nonlinear
- 7.09 Functions
- 7.10 Compare functions
- 7.11 Describe graphs qualitatively
- 7.12 Linear models

### III.8 . GEOMETRY (23)

---

- 8.01 Angle relationships
- 8.02 Parallel lines & a transversal
- 8.03 Triangle angle sum
- 8.04 Area of parallelograms
- 8.05 Area of triangles
- 8.06 Area of trapezoids
- 8.07 Composite area
- 8.08 Circle parts
- 8.09 Circumference
- 8.10 Area of a circle
- 8.11 Surface area of prisms
- 8.12 Volume of prisms
- 8.13 Volume of cylinders
- 8.14 Volume of cones & spheres
- 8.15 Pythagorean theorem
- 8.16 Converse of Pythagoras
- 8.17 Distance on the plane
- 8.18 Pythagoras in 3D
- 8.19 Translations
- 8.20 Reflections
- 8.21 Rotations
- 8.22 Dilations
- 8.23 Congruence & similarity informally

### III.9 . STATISTICS & PROBABILITY (20)

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- 9.01 Statistical questions
- 9.02 Mean
- 9.03 Median & mode
- 9.04 Range
- 9.05 Quartiles & IQR
- 9.06 Box plots

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|      |                          |
|------|--------------------------|
| 9.07 | Dot plots                |
| 9.08 | Histograms               |
| 9.09 | Mean absolute deviation  |
| 9.10 | Sampling                 |
| 9.11 | Inferences from samples  |
| 9.12 | Compare populations      |
| 9.13 | Probability basics       |
| 9.14 | Experimental probability |
| 9.15 | Sample spaces            |
| 9.16 | Compound events          |
| 9.17 | Simulations              |
| 9.18 | Scatter plots            |
| 9.19 | Trend lines              |
| 9.20 | Two-way tables           |

# Era IV . Solve - Peak Range

199 skills . 15 domains . Find the unknown.

## IV.1 . EQUATIONS & INEQUALITIES (5)

---

- 1.01 Solve one-step equations
- 1.02 Solve two-step equations
- 1.03 Solve multi-step equations
- 1.04 Solve and graph inequalities
- 1.05 Compound inequalities

## IV.2 . FUNCTIONS (15)

---

- 2.01 Function or not
- 2.02 Function notation  $f(x)$
- 2.03 Evaluate functions
- 2.04 Domain from a graph
- 2.05 Range from a graph
- 2.06 Domain from an equation
- 2.07 Vertical line test
- 2.08 Independent & dependent
- 2.09 Discrete vs continuous
- 2.10 Increasing & decreasing
- 2.11 Maxima & minima (visual)
- 2.12 Average rate of change
- 2.13 Piecewise graphs
- 2.14 Even & odd (visual)
- 2.15 Inverse from a table

## IV.3 . LINEAR FUNCTIONS (9)

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- 3.01 Slope from two points
- 3.02 Slope from a graph
- 3.03 Slope-intercept form
- 3.04 Point-slope form
- 3.05 Standard form
- 3.06 Graph from intercepts
- 3.07 Parallel & perpendicular slopes
- 3.08 Write equation from graph
- 3.09 Model with linear functions

## IV.4 . SYSTEMS (15)

---

- 4.01 Solve by graphing
- 4.02 Solve by substitution
- 4.03 Solve by elimination
- 4.04 Systems of inequalities
- 4.05 Three-variable intro
- 4.06 Consistent & inconsistent
- 4.07 Dependent systems
- 4.08 Word: mixture
- 4.09 Word: motion
- 4.10 Word: value
- 4.11 Break-even models

- 4.12 Linear programming region
- 4.13 Matrix row ops intro
- 4.14 Augmented matrices
- 4.15 Gaussian elimination (2x2)

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#### IV.5 . EXPONENTS & RADICALS (7)

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- 5.01 Product & quotient rules
- 5.02 Power rules
- 5.03 Negative exponents
- 5.04 Rational exponents
- 5.05 Simplify radicals
- 5.06 Operations with radicals
- 5.07 Rationalize denominators

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#### IV.6 . POLYNOMIALS (16)

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- 6.01 Classify by degree & terms
- 6.02 Add & subtract polynomials
- 6.03 Multiply monomials
- 6.04 FOIL & distribute
- 6.05 Special products
- 6.06 GCF factoring
- 6.07 Factor trinomials  $a=1$
- 6.08 Factor trinomials  $a \neq 1$
- 6.09 Difference of squares
- 6.10 Sum/diff of cubes
- 6.11 Factor by grouping
- 6.12 Factor completely
- 6.13 Divide by monomial
- 6.14 Long division of polynomials
- 6.15 Synthetic division
- 6.16 Remainder theorem

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#### IV.7 . QUADRATICS (17)

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- 7.01 Graph parabolas from vertex
- 7.02 Axis of symmetry
- 7.03 Intercept form
- 7.04 Vertex form conversions
- 7.05 Complete the square
- 7.06 Quadratic formula
- 7.07 Discriminant
- 7.08 Complex roots intro
- 7.09 Projectile models
- 7.10 Max/min word problems
- 7.11 Factor to solve
- 7.12 Square-root method
- 7.13 Systems: linear-quadratic
- 7.14 Quadratic inequalities
- 7.15 Transformations of  $y=x^2$
- 7.16 Focus-directrix intro
- 7.17 Fit a quadratic

**IV.8 . COMPLEX NUMBERS (16)**

---

|      |                             |
|------|-----------------------------|
| 8.01 | Imaginary unit $i$          |
| 8.02 | Powers of $i$               |
| 8.03 | Add & subtract complex      |
| 8.04 | Multiply complex            |
| 8.05 | Complex conjugates          |
| 8.06 | Divide complex              |
| 8.07 | Modulus                     |
| 8.08 | Argand plane                |
| 8.09 | Polar form                  |
| 8.10 | De Moivre (powers)          |
| 8.11 | $n$ th roots of complex     |
| 8.12 | Complex from quadratics     |
| 8.13 | Operations in polar form    |
| 8.14 | Euler's formula glimpse     |
| 8.15 | Complex conjugates & graphs |
| 8.16 | Applications of complex     |

**IV.9 . FUNCTION TOOLKIT (18)**

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|      |                                   |
|------|-----------------------------------|
| 9.01 | Parent functions gallery          |
| 9.02 | Vertical shifts                   |
| 9.03 | Horizontal shifts                 |
| 9.04 | Reflections                       |
| 9.05 | Stretches & compressions          |
| 9.06 | Compose transformations           |
| 9.07 | $f(g(x))$ composition             |
| 9.08 | Decompose functions               |
| 9.09 | One-to-one & inverses             |
| 9.10 | Find inverse algebraically        |
| 9.11 | Graph inverse                     |
| 9.12 | Restrict domain for inverse       |
| 9.13 | Absolute value graphs             |
| 9.14 | Step & piecewise build            |
| 9.15 | Combine functions (+ - $\cdot$ /) |
| 9.16 | Difference quotient               |
| 9.17 | Symmetry tests                    |
| 9.18 | End behavior (visual)             |

**IV.10 . POLYNOMIAL FUNCTIONS (14)**

---

|       |                                |
|-------|--------------------------------|
| 10.01 | End behavior & degree          |
| 10.02 | Zeros & multiplicity           |
| 10.03 | Sketch from factors            |
| 10.04 | Intermediate Value Theorem     |
| 10.05 | Rational Root Theorem          |
| 10.06 | Factor theorem applications    |
| 10.07 | Complex zeros conjugate pairs  |
| 10.08 | Descartes' rule of signs       |
| 10.09 | Upper/lower bounds             |
| 10.10 | Graph with calculus-free tools |
| 10.11 | Polynomial inequalities        |
| 10.12 | Model with polynomials         |

- 10.13 Compare polynomial models
- 10.14 Turning points bound

#### IV.11 . RATIONAL & RADICAL FUNCTIONS (14)

---

- 11.01 Simplify rational expressions
- 11.02 Multiply/divide rationals
- 11.03 Add/subtract rationals
- 11.04 Complex fractions
- 11.05 Vertical asymptotes
- 11.06 Horizontal & slant asymptotes
- 11.07 Holes in graphs
- 11.08 Graph rational functions
- 11.09 Solve rational equations
- 11.10 Rational inequalities
- 11.11 Square-root functions
- 11.12 Cube-root functions
- 11.13 Solve radical equations
- 11.14 Extraneous roots

#### IV.12 . EXPONENTIAL & LOGARITHMIC (17)

---

- 12.01 Exponential growth graphs
- 12.02 Exponential decay
- 12.03 e and continuous growth
- 12.04 Log as inverse of exp
- 12.05 Common & natural logs
- 12.06 Log properties: product
- 12.07 Log properties: quotient/power
- 12.08 Change of base
- 12.09 Solve exponential equations
- 12.10 Solve log equations
- 12.11 Exponential models (finance)
- 12.12 Half-life & doubling
- 12.13 Newton's law of cooling
- 12.14 Logarithmic scales
- 12.15 Graph log functions
- 12.16 Transform exp & log
- 12.17 Compare growth rates

#### IV.13 . SEQUENCES & SERIES (14)

---

- 13.01 Sequence notation  $a_n$
- 13.02 Arithmetic sequences
- 13.03 Arithmetic series
- 13.04 Geometric sequences
- 13.05 Geometric series (finite)
- 13.06 Infinite geometric  $|r| < 1$
- 13.07 Recursive definitions
- 13.08 Explicit from recursive
- 13.09 Sigma notation
- 13.10 Partial sums
- 13.11 Fibonacci & variants
- 13.12 Applications of series

13.13 Binomial coefficients

13.14 Binomial theorem

---

#### **IV.14 . COUNTING & PROBABILITY (12)**

---

14.01 Fundamental counting

14.02 Permutations

14.03 Combinations

14.04 Distinguishable permutations

14.05 Pascal & combinations

14.06 Sample spaces

14.07 Theoretical probability

14.08 Independent events

14.09 Dependent events

14.10 Mutually exclusive

14.11 Conditional probability

14.12 Expected value intro

---

#### **IV.15 . DATA & DISTRIBUTIONS (10)**

---

15.01 Dot plots & histograms

15.02 Mean, median, mode

15.03 Range & IQR

15.04 Box plots

15.05 Variance & SD

15.06 Z-scores

15.07 Normal curve intro

15.08 Empirical rule

15.09 Scatter & correlation

15.10 Least-squares line

# Era V . Prove - Proof Palace

173 skills . 15 domains . Know it's true.

## V.1 . LOGIC & PROOF (16)

---

- 1.01 Statements & negations
- 1.02 Conditional statements
- 1.03 Converse, inverse, contrapositive
- 1.04 Biconditionals
- 1.05 Inductive reasoning
- 1.06 Deductive reasoning
- 1.07 Laws of detachment & syllogism
- 1.08 Algebraic proof
- 1.09 Two-column proof intro
- 1.10 Paragraph proofs
- 1.11 Indirect proof (contradiction)
- 1.12 Counterexamples
- 1.13 Quantifiers ? ?
- 1.14 Set language in proofs
- 1.15 Proof planning
- 1.16 Common proof patterns

## V.2 . LINES & ANGLES (6)

---

- 2.01 Angle pairs & complements
- 2.02 Vertical angles
- 2.03 Parallel lines & transversals
- 2.04 Prove lines parallel
- 2.05 Angle sum on a line
- 2.06 Perpendicular theorems

## V.3 . TRIANGLES (14)

---

- 3.01 Triangle classification
- 3.02 Triangle angle sum
- 3.03 Exterior angle theorem
- 3.04 SSS congruence
- 3.05 SAS congruence
- 3.06 ASA & AAS
- 3.07 HL for right triangles
- 3.08 CPCTC
- 3.09 Isosceles triangle theorems
- 3.10 Triangle inequality
- 3.11 Hinge theorem
- 3.12 Medians & centroid
- 3.13 Altitude & orthocenter
- 3.14 Perp. bisectors & circumcenter

## V.4 . POLYGONS (9)

---

- 4.01 Interior angle sum
- 4.02 Exterior angles of polygons
- 4.03 Regular polygons
- 4.04 Parallelogram properties

- 4.05 Rectangle, rhombus, square
- 4.06 Trapezoids & kites
- 4.07 Prove parallelogram
- 4.08 Coordinate proof of quads
- 4.09 Area formulas review

---

**V.5 . SIMILARITY (12)**

---

- 5.01 Similar figures definition
- 5.02 AA similarity
- 5.03 SSS & SAS similarity
- 5.04 Corresponding lengths
- 5.05 Triangle proportionality
- 5.06 Angle bisector theorem
- 5.07 Similar right triangles
- 5.08 Geometric mean in right ?
- 5.09 Scale drawings
- 5.10 Indirect measurement
- 5.11 Areas of similar figures
- 5.12 Volumes of similar solids

---

**V.6 . RIGHT-TRIANGLE TRIG (14)**

---

- 6.01 Pythagorean theorem
- 6.02 Converse of Pythagoras
- 6.03 Special 45-45-90
- 6.04 Special 30-60-90
- 6.05 Sine & cosine ratios
- 6.06 Tangent ratio
- 6.07 Solve right triangles
- 6.08 Angle of elevation/depression
- 6.09 Inverse trig for angles
- 6.10 Cofunction identities
- 6.11 Pythagorean identity intro
- 6.12 Applications: navigation
- 6.13 Applications: surveying
- 6.14 Trig in coordinate plane

---

**V.7 . GENERAL TRIANGLES (9)**

---

- 7.01 Law of sines
- 7.02 Ambiguous case SSA
- 7.03 Law of cosines
- 7.04 Area:  $(1/2)ab \sin C$
- 7.05 Heron's formula
- 7.06 Solve SAS / ASA / SSA
- 7.07 Solve SSS / SAS
- 7.08 Bearing problems
- 7.09 Vectors in the plane intro

---

**V.8 . CIRCLES (17)**

---

- 8.01 Circle vocabulary
- 8.02 Central angles & arcs
- 8.03 Inscribed angles

|      |                              |
|------|------------------------------|
| 8.04 | Angles from chords           |
| 8.05 | Tangents from a point        |
| 8.06 | Tangent-chord angles         |
| 8.07 | Intersecting chords          |
| 8.08 | Secant & tangent lengths     |
| 8.09 | Arc length                   |
| 8.10 | Sector area                  |
| 8.11 | Segment area                 |
| 8.12 | Equation of a circle         |
| 8.13 | Complete square for circle   |
| 8.14 | Inscribed & circumscribed    |
| 8.15 | Cyclic quadrilaterals        |
| 8.16 | Power of a point             |
| 8.17 | Common tangent constructions |

## V.9 . TRIG FUNCTIONS (15)

---

|      |                            |
|------|----------------------------|
| 9.01 | Unit circle definition     |
| 9.02 | Radians & degrees          |
| 9.03 | Coterminal angles          |
| 9.04 | Reference angles           |
| 9.05 | Sine & cosine graphs       |
| 9.06 | Amplitude & period         |
| 9.07 | Phase shift                |
| 9.08 | Tangent graph              |
| 9.09 | Sec, csc, cot graphs       |
| 9.10 | Domain & range of trig     |
| 9.11 | Inverse sine/cosine/tan    |
| 9.12 | Solve for angles on circle |
| 9.13 | Sinusoidal models          |
| 9.14 | Harmonic motion intro      |
| 9.15 | Polar angle sense          |

## V.10 . IDENTITIES & EQUATIONS (12)

---

|       |                              |
|-------|------------------------------|
| 10.01 | Reciprocal identities        |
| 10.02 | Quotient identities          |
| 10.03 | Pythagorean identities       |
| 10.04 | Even/odd trig                |
| 10.05 | Cofunction identities        |
| 10.06 | Sum & difference formulas    |
| 10.07 | Double-angle formulas        |
| 10.08 | Half-angle formulas          |
| 10.09 | Verify identities            |
| 10.10 | Solve trig equations (basic) |
| 10.11 | Solve trig equations (multi) |
| 10.12 | Product-to-sum glimpse       |

## V.11 . COORDINATE GEOMETRY (9)

---

|       |                           |
|-------|---------------------------|
| 11.01 | Distance & midpoint       |
| 11.02 | Partition a segment       |
| 11.03 | Slope criteria proofs     |
| 11.04 | Equation of a line review |

- 11.05 Parallel/perp in coordinates
- 11.06 Coordinate triangle proofs
- 11.07 Locus problems
- 11.08 Area via coordinates
- 11.09 Transformations on plane

---

**V.12 . CONIC SECTIONS (11)**

---

- 12.01 Parabola focus-directrix
- 12.02 Parabola equations
- 12.03 Ellipse definition
- 12.04 Ellipse standard form
- 12.05 Hyperbola definition
- 12.06 Hyperbola standard form
- 12.07 Classify conics
- 12.08 Complete the square (conics)
- 12.09 Translate conics
- 12.10 Eccentricity
- 12.11 Applications of conics

---

**V.13 . TRANSFORMATIONS (12)**

---

- 13.01 Translations
- 13.02 Reflections
- 13.03 Rotations 90/180/270
- 13.04 Dilations
- 13.05 Composition of isometries
- 13.06 Glide reflections
- 13.07 Symmetry groups intro
- 13.08 Mapping figures
- 13.09 Coordinate rules
- 13.10 Invariant properties
- 13.11 Tessellations
- 13.12 Frieze patterns glimpse

---

**V.14 . SOLIDS (8)**

---

- 14.01 Prisms & cylinders
- 14.02 Pyramids & cones
- 14.03 Spheres
- 14.04 Surface area
- 14.05 Volume formulas
- 14.06 Cavalieri's principle
- 14.07 Cross sections
- 14.08 Nets & 3D reasoning

---

**V.15 . CONSTRUCTIONS (9)**

---

- 15.01 Copy a segment
- 15.02 Copy an angle
- 15.03 Perpendicular bisector
- 15.04 Angle bisector
- 15.05 Parallel through a point
- 15.06 Equilateral triangle
- 15.07 Incircle & circumcircle

- 
- 15.08 Tangent from external point
  - 15.09 Classical impossible glimpse

# Era VI . Motion - Tide Sea

174 skills . 12 domains . Understand change.

## VI.1 . LIMITS & CONTINUITY (20)

---

|      |                                        |
|------|----------------------------------------|
| 1.01 | Intuitive limit idea                   |
| 1.02 | Limit from a table                     |
| 1.03 | Limit from a graph                     |
| 1.04 | One-sided limits                       |
| 1.05 | Limit laws                             |
| 1.06 | Direct substitution                    |
| 1.07 | Factoring to find limits               |
| 1.08 | Rationalize for limits                 |
| 1.09 | Limits at infinity                     |
| 1.10 | Infinite limits & asymptotes           |
| 1.11 | Squeeze theorem                        |
| 1.12 | Trig limits $(\sin x)/x$               |
| 1.13 | Continuity at a point                  |
| 1.14 | Types of discontinuity                 |
| 1.15 | IVT applications                       |
| 1.16 | Limits of piecewise                    |
| 1.17 | $\epsilon$ - $\delta$ definition intro |
| 1.18 | Limit of a sequence                    |
| 1.19 | Unbounded behavior                     |
| 1.20 | Comparing growth (limits)              |

## VI.2 . DERIVATIVES (18)

---

|      |                                 |
|------|---------------------------------|
| 2.01 | Tangent slope idea              |
| 2.02 | Definition of derivative        |
| 2.03 | Derivative as function          |
| 2.04 | Differentiability vs continuity |
| 2.05 | Power rule                      |
| 2.06 | Constant & sum rules            |
| 2.07 | Product rule                    |
| 2.08 | Quotient rule                   |
| 2.09 | Chain rule                      |
| 2.10 | Derivatives of trig             |
| 2.11 | Derivatives of exp & log        |
| 2.12 | Implicit differentiation        |
| 2.13 | Higher-order derivatives        |
| 2.14 | Derivative from graph           |
| 2.15 | Numerical derivatives           |
| 2.16 | Logarithmic differentiation     |
| 2.17 | Inverse function derivatives    |
| 2.18 | Related rates setup             |

## VI.3 . APPLICATIONS OF DERIVATIVES (21)

---

|      |                           |
|------|---------------------------|
| 3.01 | Position, velocity, accel |
| 3.02 | Related rates problems    |
| 3.03 | Linearization             |
| 3.04 | Differentials             |

|      |                             |
|------|-----------------------------|
| 3.05 | Extreme Value Theorem       |
| 3.06 | Critical points             |
| 3.07 | Closed interval method      |
| 3.08 | Mean Value Theorem          |
| 3.09 | Rolle's theorem             |
| 3.10 | First derivative test       |
| 3.11 | Second derivative test      |
| 3.12 | Concavity & inflection      |
| 3.13 | Curve sketching             |
| 3.14 | Optimization (geometry)     |
| 3.15 | Optimization (business)     |
| 3.16 | Newton's method             |
| 3.17 | L'Hôpital's rule            |
| 3.18 | Indeterminate forms         |
| 3.19 | Motion along a line         |
| 3.20 | Marginal analysis           |
| 3.21 | Summary: $f$ , $f'$ , $f''$ |

#### VI.4 . INTEGRALS (15)

---

|      |                              |
|------|------------------------------|
| 4.01 | Area under a curve idea      |
| 4.02 | Riemann sums                 |
| 4.03 | Definite integral definition |
| 4.04 | Midpoint & trapezoid rules   |
| 4.05 | Antiderivatives              |
| 4.06 | Indefinite integrals         |
| 4.07 | Fundamental Theorem I        |
| 4.08 | Fundamental Theorem II       |
| 4.09 | Net change theorem           |
| 4.10 | Substitution (u-sub)         |
| 4.11 | Integrals of trig            |
| 4.12 | Integrals of exp & log       |
| 4.13 | Even & odd integrands        |
| 4.14 | Average value of a function  |
| 4.15 | Accumulation functions       |

#### VI.5 . INTEGRATION TECHNIQUES (16)

---

|      |                                 |
|------|---------------------------------|
| 5.01 | Integration by parts            |
| 5.02 | Trig integrals (powers)         |
| 5.03 | Trig substitution               |
| 5.04 | Partial fractions (linear)      |
| 5.05 | Partial fractions (quadratic)   |
| 5.06 | Improper integrals (?)          |
| 5.07 | Improper integrals (discont.)   |
| 5.08 | Completing square in integrals  |
| 5.09 | Rationalizing substitutions     |
| 5.10 | Tables of integrals             |
| 5.11 | Numerical integration error     |
| 5.12 | Strategy: which method?         |
| 5.13 | Reduction formulas glimpse      |
| 5.14 | Integrals yielding inverse trig |
| 5.15 | Hyperbolic intro (optional)     |

5.16 CAS vs by-hand judgment

---

## VI.6 . APPLICATIONS OF INTEGRALS (15)

---

- 6.01 Area between curves
- 6.02 Disk method
- 6.03 Washer method
- 6.04 Shell method
- 6.05 Arc length
- 6.06 Surface of revolution
- 6.07 Work (springs, pumps)
- 6.08 Hydrostatic force
- 6.09 Moments & center of mass
- 6.10 Probability density intro
- 6.11 Present & future value
- 6.12 Biology: cardiac output
- 6.13 Consumer/producer surplus
- 6.14 Volume by known cross-section
- 6.15 Compare volume methods

---

## VI.7 . DIFFERENTIAL EQUATIONS I (12)

---

- 7.01 What is a DE?
- 7.02 Solution verification
- 7.03 Direction fields
- 7.04 Euler's method
- 7.05 Separable equations
- 7.06 Exponential growth/decay DE
- 7.07 Newton's law of cooling DE
- 7.08 Logistic equation
- 7.09 Mixing problems
- 7.10 First-order linear (?)
- 7.11 Existence & uniqueness idea
- 7.12 Autonomous equations

---

## VI.8 . PARAMETRIC & POLAR (12)

---

- 8.01 Parametric equations
- 8.02 Eliminate the parameter
- 8.03 Derivatives  $dy/dx$  parametric
- 8.04 Arc length parametric
- 8.05 Polar coordinates
- 8.06 Convert polar ? cartesian
- 8.07 Graph polar roses & limaçons
- 8.08 Tangents in polar
- 8.09 Area in polar
- 8.10 Arc length in polar
- 8.11 Conics in polar
- 8.12 Parametric motion apps

---

## VI.9 . SEQUENCES & SERIES (11)

---

- 9.01 Sequence convergence
- 9.02 Monotone & bounded
- 9.03 Series & partial sums

|      |                         |
|------|-------------------------|
| 9.04 | Geometric series review |
| 9.05 | Telescoping series      |
| 9.06 | Divergence test         |
| 9.07 | Integral test           |
| 9.08 | Comparison tests        |
| 9.09 | Alternating series      |
| 9.10 | Ratio & root tests      |
| 9.11 | Absolute vs conditional |

---

## VI.10 . POWER & TAYLOR SERIES (16)

---

|       |                            |
|-------|----------------------------|
| 10.01 | Power series idea          |
| 10.02 | Radius of convergence      |
| 10.03 | Interval of convergence    |
| 10.04 | Differentiate power series |
| 10.05 | Integrate power series     |
| 10.06 | Taylor polynomials         |
| 10.07 | Taylor & Maclaurin series  |
| 10.08 | Common Maclaurin series    |
| 10.09 | Binomial series            |
| 10.10 | Remainder estimates        |
| 10.11 | Approximate integrals      |
| 10.12 | Solving DE with series     |
| 10.13 | Analytic functions glimpse |
| 10.14 | Fourier vs Taylor contrast |
| 10.15 | Applications: constants    |
| 10.16 | Power series manipulations |

---

## VI.11 . STATISTICAL INFERENCE (9)

---

|       |                              |
|-------|------------------------------|
| 11.01 | Sampling distributions       |
| 11.02 | CLT idea                     |
| 11.03 | Confidence intervals (mean)  |
| 11.04 | Confidence intervals (prop.) |
| 11.05 | Hypothesis test logic        |
| 11.06 | p-values                     |
| 11.07 | t-procedures                 |
| 11.08 | Chi-square intro             |
| 11.09 | Type I / Type II errors      |

---

## VI.12 . MECHANICS (9)

---

|       |                             |
|-------|-----------------------------|
| 12.01 | Displacement from velocity  |
| 12.02 | Work-energy glimpse         |
| 12.03 | Hooke's law & springs       |
| 12.04 | Center of mass (calc)       |
| 12.05 | Projectile via calc         |
| 12.06 | Simple harmonic from DE     |
| 12.07 | Escape velocity idea        |
| 12.08 | Kepler glimpse              |
| 12.09 | Force & acceleration models |

# Era VII . Space - Star Vault

197 skills . 19 domains . Explore higher worlds.

## VII.1 . VECTORS IN $R^n$ (12)

---

- 1.01 Vector geometry in  $R^2/R^3$
- 1.02 Vector addition & scaling
- 1.03 Dot product
- 1.04 Orthogonality
- 1.05 Projections
- 1.06 Cross product
- 1.07 Lines in  $R^3$
- 1.08 Planes in  $R^3$
- 1.09 Distances: point-line-plane
- 1.10 Linear combinations
- 1.11 Span of vectors
- 1.12 Norm & unit vectors

## VII.2 . LINEAR SYSTEMS (9)

---

- 2.01 Row echelon form
- 2.02 Reduced row echelon
- 2.03 Gaussian elimination
- 2.04 Free variables & params
- 2.05 Homogeneous systems
- 2.06 Consistency criteria
- 2.07 Rank idea
- 2.08 Applications: networks
- 2.09 Applications: balance

## VII.3 . MATRIX ALGEBRA (12)

---

- 3.01 Matrix operations
- 3.02 Matrix multiplication
- 3.03 Identity & inverse
- 3.04 Elementary matrices
- 3.05 Solve via inverse
- 3.06 Transpose
- 3.07 Block matrices
- 3.08 Matrix powers
- 3.09 LU factorization intro
- 3.10 Sparse vs dense
- 3.11 Computer arithmetic caution
- 3.12 Affine maps as matrices

## VII.4 . VECTOR SPACES (16)

---

- 4.01 Axioms of a vector space
- 4.02 Subspaces
- 4.03 Null space
- 4.04 Column & row space
- 4.05 Basis
- 4.06 Dimension
- 4.07 Coordinates in a basis

- 4.08 Change of basis
- 4.09 Rank-nullity theorem
- 4.10 Examples:  $P_n$ ,  $C[a,b]$
- 4.11 Affine subspaces
- 4.12 Direct sums
- 4.13 Isomorphism of spaces
- 4.14 Standard bases
- 4.15 Extending independent sets
- 4.16 Extracting bases from spans

## VII.5 . LINEAR TRANSFORMATIONS (9)

---

- 5.01 Definition & examples
- 5.02 Matrix of a linear map
- 5.03 Kernel & image
- 5.04 Composition of maps
- 5.05 Invertible transformations
- 5.06 Similarity of matrices
- 5.07 Projections as maps
- 5.08 Rotations & reflections
- 5.09 Rank of a transformation

## VII.6 . DETERMINANTS (9)

---

- 6.01 2x2 and 3x3 determinants
- 6.02 Cofactor expansion
- 6.03 Properties of det
- 6.04 Det & invertibility
- 6.05 Det as volume
- 6.06 Cramer's rule
- 6.07 Adjugate & inverse
- 6.08 Det under row ops
- 6.09 Orientation & sign

## VII.7 . EIGENVALUES & EIGENVECTORS (9)

---

- 7.01 Eigenvalue equation
- 7.02 Characteristic polynomial
- 7.03 Eigenspaces
- 7.04 Diagonalization
- 7.05 When diagonalizable?
- 7.06 Powers via diagonalization
- 7.07 Markov chains intro
- 7.08 Differential systems preview
- 7.09 Complex eigenvalues

## VII.8 . ORTHOGONALITY (12)

---

- 8.01 Orthogonal sets
- 8.02 Orthonormal bases
- 8.03 Gram-Schmidt
- 8.04 QR factorization
- 8.05 Orthogonal projections
- 8.06 Least squares
- 8.07 Normal equations

- 8.08 Orthogonal matrices
- 8.09 Fourier via ortho glimpse
- 8.10 Cauchy-Schwarz
- 8.11 Triangle inequality (norms)
- 8.12 Best approximation

---

## VII.9 . SYMMETRIC MATRICES & SVD (8)

---

- 9.01 Spectral theorem (sym)
- 9.02 Orthogonal diagonalization
- 9.03 Quadratic forms
- 9.04 Positive definite
- 9.05 SVD idea
- 9.06 Singular values
- 9.07 Low-rank approximation
- 9.08 PCA glimpse

---

## VII.10 . FUNCTIONS OF SEVERAL VARIABLES (6)

---

- 10.01 Graphs & level curves
- 10.02 Level surfaces
- 10.03 Limits in  $\mathbb{R}^2$
- 10.04 Continuity in  $\mathbb{R}^2$
- 10.05 Domain in several variables
- 10.06 Quadric surfaces review

---

## VII.11 . PARTIAL DERIVATIVES (12)

---

- 11.01 Partial derivatives
- 11.02 Higher partials & Clairaut
- 11.03 Tangent planes
- 11.04 Linear approximation (multi)
- 11.05 Chain rule (multi)
- 11.06 Directional derivatives
- 11.07 Gradient vector
- 11.08 Gradient & level sets
- 11.09 Implicit differentiation (multi)
- 11.10 Differentiability in  $\mathbb{R}^n$
- 11.11 Jacobian matrix
- 11.12 Taylor in two variables

---

## VII.12 . OPTIMIZATION (8)

---

- 12.01 Critical points in  $\mathbb{R}^2$
- 12.02 Second derivative test (multi)
- 12.03 Absolute extrema on regions
- 12.04 Lagrange multipliers
- 12.05 Constrained optimization apps
- 12.06 Least squares as opt
- 12.07 Gradient descent glimpse
- 12.08 Convexity intro

---

**VII.13 . MULTIPLE INTEGRALS (9)**

---

- 13.01 Double integrals over rectangles
- 13.02 Iterated integrals
- 13.03 Fubini's theorem
- 13.04 Double integrals over regions
- 13.05 Polar double integrals
- 13.06 Applications: mass, center
- 13.07 Triple integrals
- 13.08 Cylindrical coordinates
- 13.09 Spherical coordinates

---

**VII.14 . VECTOR CALCULUS (18)**

---

- 14.01 Vector fields
- 14.02 Line integrals (scalar)
- 14.03 Line integrals (vector)
- 14.04 Conservative fields
- 14.05 Fundamental theorem for lines
- 14.06 Green's theorem
- 14.07 Curl & divergence
- 14.08 Parametric surfaces
- 14.09 Surface integrals (scalar)
- 14.10 Flux integrals
- 14.11 Stokes' theorem
- 14.12 Divergence theorem
- 14.13 Path independence
- 14.14 Potential functions
- 14.15 Applications: work & flow
- 14.16 Maxwell glimpse
- 14.17 Differential forms taste
- 14.18 When theorems apply

---

**VII.15 . FIRST-ORDER EQUATIONS (7)**

---

- 15.01 Review separable & linear
- 15.02 Exact equations
- 15.03 Integrating factors (exact)
- 15.04 Homogeneous first-order
- 15.05 Bernoulli equations
- 15.06 Substitutions
- 15.07 Qualitative analysis

---

**VII.16 . SECOND-ORDER LINEAR (13)**

---

- 16.01 Linear 2nd-order structure
- 16.02 Homogeneous constant coeff
- 16.03 Characteristic equation
- 16.04 Repeated & complex roots
- 16.05 Nonhomogeneous: undetermined
- 16.06 Variation of parameters
- 16.07 Spring-mass systems
- 16.08 Damping regimes
- 16.09 Forced oscillations
- 16.10 Resonance

- 16.11 Cauchy-Euler equations
- 16.12 Boundary value intro
- 16.13 Reduction of order

---

**VII.17 . SYSTEMS OF ODES (11)**

---

- 17.01 First-order systems
- 17.02 Convert higher-order to system
- 17.03 Matrix form  $\mathbf{x}' = \mathbf{A}\mathbf{x}$
- 17.04 Eigen-method for systems
- 17.05 Phase portraits (2D)
- 17.06 Spiral, node, saddle
- 17.07 Nonhomogeneous systems
- 17.08 Linearization of nonlinear
- 17.09 Predator-prey glimpse
- 17.10 Coupled springs
- 17.11 Stability language

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**VII.18 . LAPLACE TRANSFORM (9)**

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- 18.01 Definition & existence
- 18.02 Transforms of basics
- 18.03 s-shift & t-shift
- 18.04 Solve IVP with Laplace
- 18.05 Step & impulse functions
- 18.06 Convolution
- 18.07 Transfer functions glimpse
- 18.08 Partial fractions (Laplace)
- 18.09 Systems via Laplace

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**VII.19 . FOURIER SERIES (8)**

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- 19.01 Periodic functions
- 19.02 Fourier coefficients
- 19.03 Sine & cosine series
- 19.04 Convergence idea
- 19.05 Even/odd extensions
- 19.06 Half-range expansions
- 19.07 Parseval glimpse
- 19.08 PDE motivation (heat/wave)